Ruh.ai V1 Tenant Overlay Kit

Construction Project Operations - Procore + SharePoint + Teams

Deployment-ready overlay for one contractor, one trust boundary, and repeatable project operations workflows

VERSION	PACK REF	PRIMARY USE	DATE
0.1.0	construction-project-operations@1.0.0	Pilot + first production overlay	2026-03-08

Executive snapshot

This overlay kit binds the reusable Construction Project Operations Pack to a contractor stack where Procore is the official project record, SharePoint stores durable documents and generated outputs, and Microsoft Teams is the collaboration and approval surface.

The overlay is designed for general contractors and larger specialty contractors that already operate one project at a time through structured PMIS and document systems but still spend too much staff time on routing, document assembly, draft preparation, follow-up, and weekly reporting.

The recommended first activation slice is intake and routing, RFI lifecycle, submittal lifecycle, and meeting-action sync.

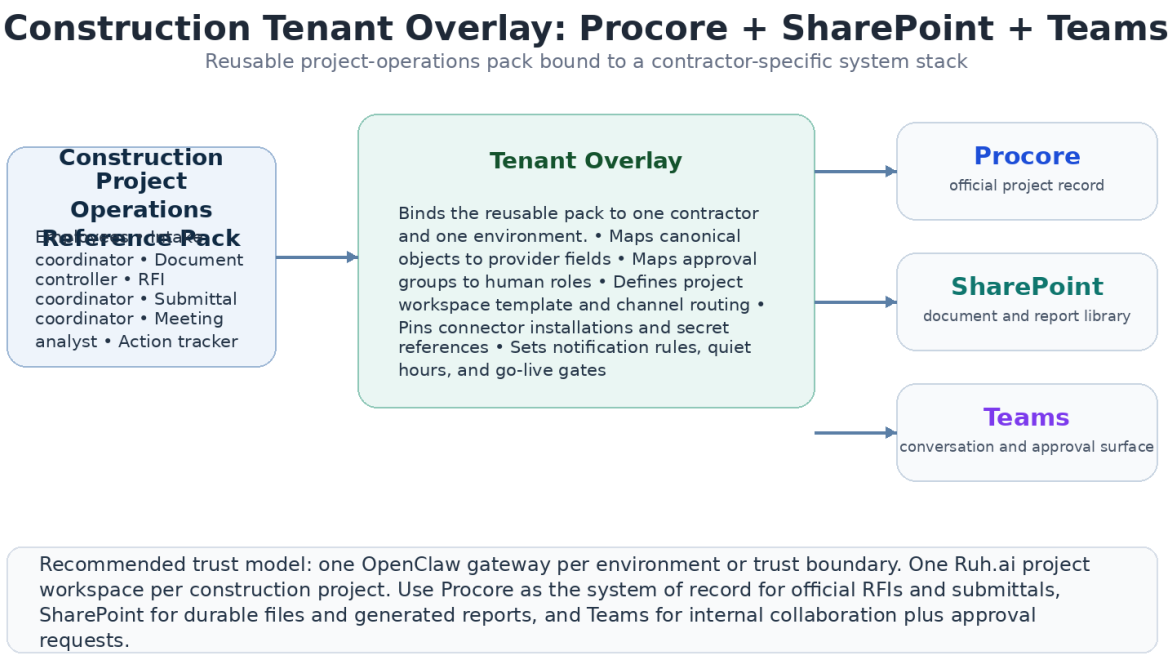

Figure 1. The overlay binds one reusable pack to one contractor stack and one trust boundary.

1. Purpose, scope, and design choices

This document is the deployment companion to the Construction Project Operations reference pack. It translates the horizontal pack into a concrete tenant-specific configuration pattern for contractors running Procore, SharePoint, and Teams. The kit defines how canonical construction objects are mapped, which employees are enabled, where official records live, what approval rules apply, and what must be true before the overlay can be promoted from pilot to production.

- Use Procore as the official record for RFIs, submittals, and project metadata that already live in the PMIS.
- Use SharePoint as the durable repository for generated meeting packs, status reports, exported bundles, and supporting project files that should remain easily shareable inside Microsoft 365.
- Use Teams as the internal collaboration surface for notifications, approvals, and daily operator interaction with digital employees.
- Keep one Ruh.ai project workspace per construction project so context, citations, and approvals stay cleanly scoped.

Out of scope in this overlay

Autonomous safety decisions, schedule resequencing, financial commitments, procurement commitments, or any action that changes project obligations without human approval.
Replacing Procore, SharePoint, or Teams. The overlay sits above them and writes back into them in a controlled way.

2. Stack assumptions and tenant prerequisites

The overlay only works cleanly when the contractor already has one digital backbone per project. The goal is not to digitize a paper-first organization from scratch; the goal is to remove administrative drag from a digitally active team.

Area	Minimum requirement	Why it matters
Procore	API-enabled company account, project list, RFIs and submittals already managed in Procore	The overlay depends on stable IDs and official write-back targets.
SharePoint	Project libraries with controlled access and consistent folder patterns	Generated packs and citation bundles need a durable document home.
Teams	Project or department channels plus an approval channel for operations leads	Human-in-the-loop approvals and digest delivery need a reliable surface.
Identity	Named users and groups for project engineers, document control, PMs, and escalations	Policies must resolve to real approvers instead of free-text names.
Project coding	Consistent project code / project name convention across systems	Cross-system routing is much easier when one project key exists everywhere.

3. Overlay topology and trust-boundary model

The overlay is applied at the tenant environment level, then parameterized per project. The recommended topology is one OpenClaw gateway for each trust boundary such as dev, staging, and production, with one Ruh.ai project workspace inside that boundary for each active construction project. This matches OpenClaw's documented one-trust-boundary-per-gateway model and keeps project memory separated from other projects.

- Tenant level: contractor-wide defaults, connector installations, approval role mappings, and notification defaults.
- Environment level: dev, staging, and production runtime settings, secret references, and gateway profile.
- Project level: Procore project ID, SharePoint library path, Teams channel bindings, project contacts, and workflow activation state.
- Workspace level: citation cache, generated drafts, temporary bundles, and run telemetry for one project only.

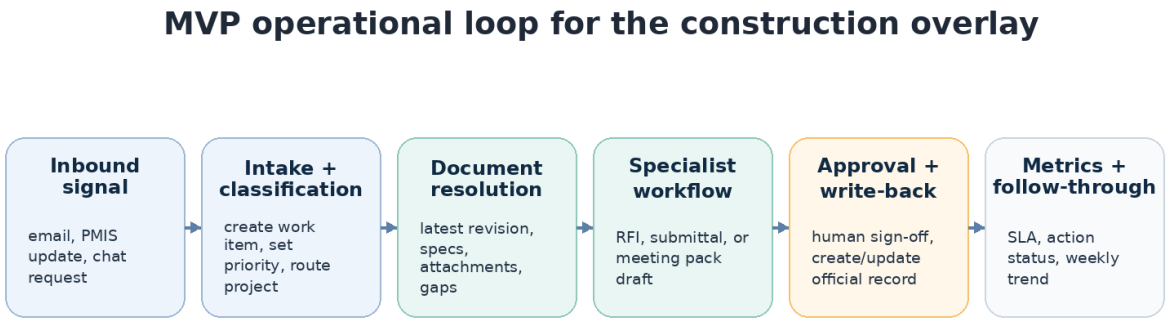

Figure 2. MVP workflow loop recommended for the first construction overlay rollout.

4. Canonical object mapping

The overlay maps provider-specific fields to the pack's canonical construction vocabulary. Workflow logic should consume canonical objects; only the connector layer should know vendor field names.

Canonical object	Primary source	Secondary source	Notes
Project	Procore project	Teams channel prefix + SharePoint library	Project code should be the stable cross-system key.
DocumentRef	SharePoint file or Procore attachment	SharePoint controlled file path	Document controller resolves the latest approved version.
RFI	Procore RFI record	SharePoint citation bundle copy	Official state stays in Procore.
Submittal	Procore submittal package	SharePoint supporting docs	Official review state stays in Procore.
MeetingPacket	SharePoint generated file	Teams delivery message	Teams distributes; SharePoint stores.
ActionItem	Ruh.ai task artifact	Teams follow-up thread	Action tracker owns due-date reminders and status rollups.
ApprovalGroup	Azure AD / M365 group	Ruh.ai policy reference	No free-text approver resolution in production.

5. Employee activation and permission envelope

The tenant overlay should never enable a digital employee without an explicit permission envelope. The matrix below is the minimum v1 posture.

Employee	Primary actions	Auto allowed	Approval required
Project intake coordinator	Classify inbound requests, open work items, route to project	Create internal tasks and Teams summaries	None by default
Document controller	Resolve latest documents and build citation bundles	Read docs, assemble context bundle	Publishing externally shared bundles
RFI coordinator	Draft RFIs and propose routing	Internal draft generation	Create or update official Procore RFI
Submittal coordinator	Prepare review package and status summary	Internal package draft, due-date watchlist	Create/update official Procore submittal record
Meeting & reporting analyst	Generate OAC pack, weekly summary, trend view	Generate SharePoint files and internal Teams posts	External distribution beyond internal channel
Action & risk tracker	Reminder nudges, overdue watch, risk status rollup	Internal reminders and status digests	Escalation outside mapped Teams channels

6. Workflow activation plan

The overlay should not activate every pack workflow on day one. Start with high-frequency, low-ambiguity flows that have a clean system of record and measurable cycle-time impact.

Workflow	Trigger	System of record	Primary metric
Intake & routing	New email, Teams request, or monitored PMIS event	Ruh.ai work item + Teams notification	Time from inbound signal to assigned task
RFI lifecycle	Question or missing-info request tied to a project	Procore RFI	Draft-to-submission turnaround and citation accuracy
Submittal lifecycle	Incoming submittal package or due-date watch condition	Procore submittal record	Cycle time and overdue reduction
Meeting-action sync	Scheduled weekly/OAC cadence or manual request	SharePoint file + Ruh.ai action items	Hours saved per week and overdue action reduction

Recommended pilot order

Week 1-2: turn on intake routing and document resolution only, with no external write-back.
Week 3-4: add supervised RFI drafting and PM approval before Procore create/update.
Week 5-6: add submittal package preparation and due-date watchlists.
Week 7-8: add recurring meeting pack generation and action tracking digests.

7. Connector profile and secrets inventory

Connector	Required access	Secret reference pattern	Health check

Procore	Read projects, RFIs, submittals; create/update permitted records	secretref://connectors/procore/*	Project list fetch + create draft dry run
SharePoint	Read controlled docs; write generated outputs to approved libraries	secretref://connectors/sharepoint/*	Library list + test write to Ruh Output
Teams	Post in mapped channels; receive bot mentions or app events	secretref://connectors/teams/*	Post test message + approval card round-trip
Ruh runtime	Bundle apply, task execution, telemetry upload	secretref://runtime/*	Adapter heartbeat + bundle checksum verification

All secrets are referenced indirectly through secret refs. No tenant overlay checked into source control should ever contain raw tokens, client secrets, or user passwords.

8. Policy and approval overlay

This overlay uses a three-tier action model to keep automation valuable without putting official project records at risk.

Risk tier	Examples	Default policy	Approver
Low	Internal summaries, task creation, reminder nudges	Auto allowed	None
Medium	Creating or updating RFI/submittal drafts in Procore	Approval required	Mapped PM or document-control role
High	Owner/designer messages, obligation-changing records, external distribution	Human only or approval required with named role	PM + escalation role if required

- No outbound message to owner, designer, or subcontractor should auto-send from the overlay in v1.
- Any missing citation bundle, stale revision warning, or document conflict automatically downgrades the action to manual review.
- Override decisions must record the approving user, timestamp, and reason code.

9. Notification, routing, and record ownership rules

Artifact / event	Primary destination	Owner of record
Task assignment	Teams project-ops channel	Ruh.ai work item/task record
Approval request	Teams approvals channel or mapped PM chat	Ruh.ai approval record
Official RFI/submittal update	Procore + Teams confirmation	Procore
Meeting packet	SharePoint file + Teams message link	SharePoint file
Daily digest	Teams project-controls channel	Ruh.ai generated summary artifact

Teams is the interaction surface, not the system of record. The overlay should send links back to Procore or SharePoint rather than treating chat history as the official audit trail.

10. Pilot acceptance gates and success metrics

Gate	Target before go-live
Connector auth	All three external connectors pass health check and least-privilege review.
Project routing	At least 95% of inbound signals route to the correct project in eval replay.
Citation quality	At least 90% of pilot drafts include valid latest-revision citations.
Approval compliance	100% of medium/high-risk actions create an approval record or remain blocked.
Cycle-time impact	Pilot shows measurable reduction in RFI/submittal administrative turnaround.

Recommended pilot KPIs

RFI draft turnaround time
Submittal package prep time
Overdue action item count
Hours saved per project engineer per week
Percent of write-backs completed only after correct approval path

11. Example overlay excerpt

The full starter file is provided separately with this kit. The excerpt below shows the intended structure and the major fields solution engineers are expected to fill.

```
kind: tenantOverlay
apiVersion: ruh.ai/v1alpha1
metadata:
  overlayId: construction.procore-sharepoint-teams
  packRef: construction-project-operations@1.0.0
  displayName: Procore + SharePoint + Teams overlay
  owner: solutions-engineering
  environmentClass: production
  version: 0.1.0
trustBoundary:
  key: contractor-prod-east
  runtimeProfile: prod
  gateway:
    mode: dedicated
    workspaceStrategy: project-per-workspace
    sandbox:
      mode: non-main
      scope: agent
projects:
  template:
    workspaceIdPattern: constr-{project_code}
    channelBindings:
      internalTeam: teams:{project_code}-ops
      approvals: teams:{project_code}-approvals
connectors:
  procore:
    enabled: true
    installationRef: secretref://connectors/procore/app-prod
    companyId: ${PROCORE_COMPANY_ID}
    projectFieldMap:
```

```
      projectId: procore.project.id
      rfiId: procore.rfi.id
      submittalId: procore.submittal.id
  sharepoint:
    enabled: true
    siteRef: ${SP_SITE_URL}
    libraryMap:
      controlledDocs: /Shared Documents/Controlled Project Files
      generatedReports: /Shared Documents/Ruh Output
  teams:
    enabled: true
    botRef: secretref://connectors/teams/bot-prod
    quietHours: "20:00-06:00 America/Chicago"
roleMappings:
  approvals:
    rfiOwner: group://project-engineers
```

12. Day-2 operations and change control

- Any change to provider mappings, approval groups, or write-back permissions should go through the same promotion flow as a code change: dev -> staging -> production.
- Project-template changes should be versioned. Never mutate active production overlays in place without a rollback snapshot.
- Connector credential rotation should be tested in staging before production cutover whenever possible.
- Escaped production mistakes should become new eval cases before the next overlay release.

Appendix A. Source assumptions

This overlay kit assumes OpenClaw is used as a trust-bounded runtime plane with isolated agents, workspaces, routing bindings, AgentSkills-compatible skill folders, configurable hooks/cron automation, and optional sandboxing. The overlay design also assumes the documented security posture of one trusted operator boundary per gateway, which is why production deployments should separate customers and environments at the gateway layer.

- [R1] Agents - OpenClaw (isolated agents, workspaces, routing).
- [R2] Skills - OpenClaw (AgentSkills-compatible skill folders and precedence).
- [R3] Hooks - OpenClaw (event-driven automations discovered from directories).
- [R4] Cron Jobs - OpenClaw (persistent gateway scheduler).
- [R5] Security - OpenClaw (one trusted operator boundary per gateway).
- [R6] Agent Workspace - OpenClaw (workspace as default cwd; sandbox recommended for isolation).